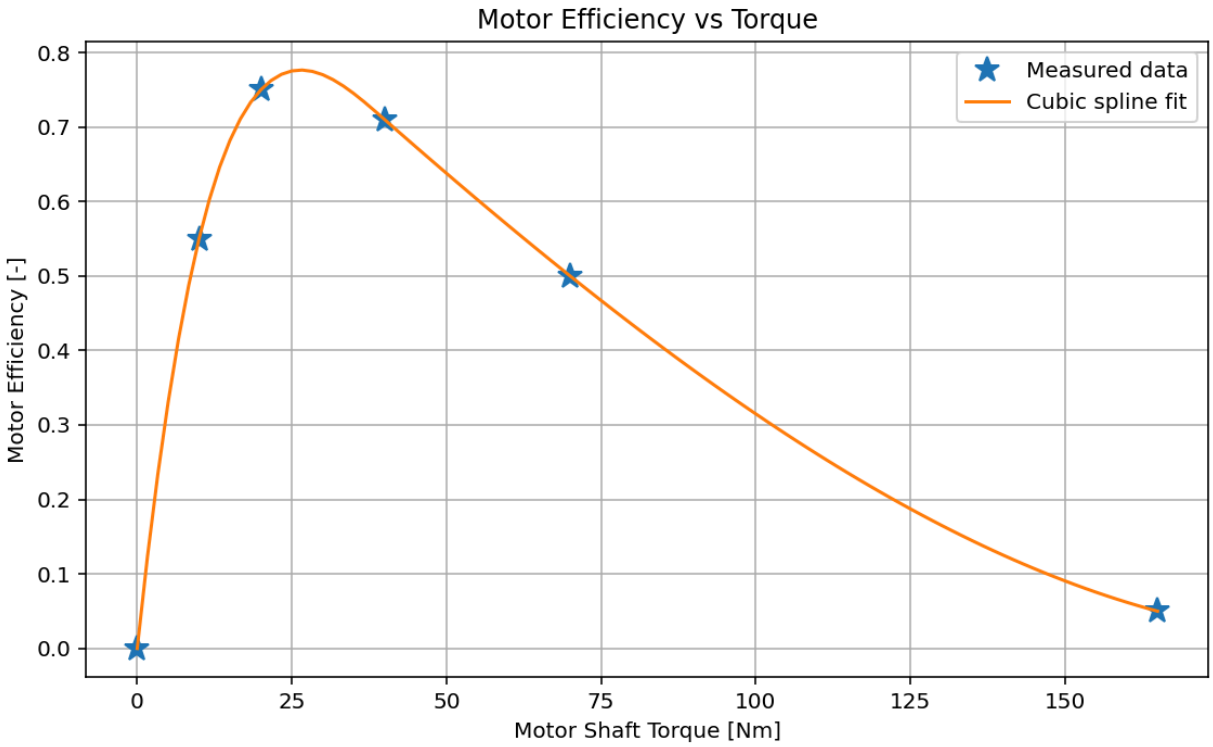


The curve created is very similar to Figure 2 of the handout, as it has the same qualitative shape. The efficiency starts at zero, rises to a peak, and then falls back toward zero at high torque. The efficiency never reaches 100 percent because there is always energy lost to heat. The curve is also smooth because cubic spline interpolation provides continuity in the first and second derivatives. Unlike linear interpolation, which produces straight segments with abrupt slope changes at each data point, cubic splines create a smooth curve without kinks or corners. This smoothness is achieved by assigning each interval its own cubic polynomial and solving the coefficients with matching-derivative conditions at shared points.

Additionally, the spline interpolation passes through all the known data points by construction. Visually, each star sits directly on the curve, confirming that the line passes through all six known data points.